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GUIDELINE ON MINIMUM REQUIREMENTS FOR INSPECTING GROUP III AND GROUP IV HAZARDOUS SUBSTANCES

This guideline delineates the mandatory regulatory framework of the Radiation Control Inspectorate sub-unit including the inspection requirements to all Group III and Group IV Hazardous Substances, in terms of the Hazardous Substances Act No. 15 of 1973 and Regulations R1332 and R247.

The guideline further provides directions to all holders of Group III and Group IV Hazardous Substances on maintaining compliance with the applicable provisions of the Act and relevant regulations. It sets out the minimum requirements and standard to be adhered to during routine, investigative, and enforcement inspections.

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DR BOITUMELO SEMETE-MAKOKOTLELA
CHIEF EXECUTIVE OFFICER

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Glossary

Abbreviation/ Term	Meaning
Act	Hazardous Substances Act No. 15 of 1973
ALARA	As Low As Reasonably Achievable
ARPO	Acting Radiation Protection Officer
Group III	Ionizing or non-ionizing electro-magnetic radiation product / device or any sonic, infrasonic and ultrasonic wave which is emitted from an electronic product
Group IV	Radioactive material which is outside a nuclear installation as defined in the Nuclear Energy Act, 1999 and which is used or intended to be used for medical, scientific, agricultural, commercial or industrial purposes, and any radioactive waste arising from such radioactive material
High-risk	Practices involving sources or equipment that have the potential to cause significant deterministic health effects if not properly controlled.
Low risk	Practices involving sources or equipment with low radiological hazard and minimal potential for significant exposure.
Moderate risk	Practices involving sources or equipment that could result in moderate radiological consequences under abnormal conditions.
R 1302	Regulation related listed electronic products
R 690	Regulation related to import of Group III hazardous substances
R1332	Regulation concerning the control of electronic products
R246	Regulation related to exclusion and exemptions
R247	Regulations related to Group IV Hazardous Substances Act
RPO	Radiation Protection Officer
SAHPRA	South African Health products Regulatory Authority

1. INTRODUCTION

The South African Health Products Regulatory Authority (SAHPRA) is mandated, in terms of the Hazardous Substances Act No. 15 of 1973 as amended, to regulate, control, and monitor the production, acquisition, importation, exportation, possession, disposal, use, operation, letting, sale, and installation of Group III and Group IV hazardous substances, in order to protect workers, the public, and the environment from radiation-related harm.

Group III and Group IV hazardous substances outline varying levels of radiological risk depending on the nature of the radioactive material, the activity involved, and the way they are used. Effective inspection of facilities and practices involving these substances through a graded regulatory approach is therefore essential to ensure compliance with regulatory requirements and adherence to radiation safety principles.

This guideline provides a standardised framework, consistent with SAHPRA's regulatory approach, promoting uniform, transparent, and effective inspections of Group III and Group IV hazardous substances.

1.1 Purpose

The purpose of this guideline is to:

- Define the minimum requirements for conducting inspections of facilities and activities involving Group III and Group IV hazardous substances.
- Provide guidance to SAHPRA inspectors on the inspection process, focus areas, and compliance expectations.
- Promote a risk-based and consistent inspection approach aligned with SAHPRA's Radiation Control framework.
- The goal is to ensure the licence or authority holders understand the entire process and are fully prepared to meet all requirements.

1.2 Scope

This guideline applies to:

- All facilities, institutions, and users authorized or required to be authorized to possess, use, store, transport, or dispose of Group III and Group IV hazardous substances.

The guideline covers routine, announced, unannounced, follow-up, and compliance inspections but does

not replace the requirements set out in legislation, licensing or authorisation conditions, or applicable SAHPRA policies.

2. LEGAL PROVISION

This guideline is issued in terms of the following legal instruments:

- Hazardous Substances Act No. 15 of 1973, as amended,
- Regulations promulgated under the Hazardous Substances Act relating to Group III and Group IV hazardous substances,
- Conditions of licencing/ authorization issued by SAHPRA, and
- Applicable national standards, codes of practice, and international guidance adopted or recognized by SAHPRA.

Compliance with this guideline does not exempt any License or Authority Holder from complying with the Act, regulations, or specific licensing or authorization conditions. Where there is a conflict between this guideline and the Act, regulations, licensing or authorization conditions, the latter shall prevail.

3. RADIATION CONTROL INSPECTION PROCESSES

Radiation Control inspections shall be conducted by SAHPRA Radiation Inspector appointed in accordance with Hazardous Substance Act, section 8 and shall follow a graded, risk-informed approach.

3.1 Pre-Inspection Activities

- Review of the licence or written authority, scope, and conditions.
- Review of previous inspection findings, non-compliances, and corrective actions.
- Assessment of the radiological risk profile, type of practice, and compliance history.
- Determination of inspection frequency and inspection scope based on the graded approach.

3.2 On-Site Inspection

- Opening meeting with management and the Radiation Protection Officer (RPO) or responsible person.
- Verification of compliance with legislative and licencing/ authorization requirements.

- Inspection of facilities, equipment, storage areas, and safety systems.
- Review of documentation, records, and training evidence.
- Interviews with relevant staff involved in the use or supervision of hazardous substances.
- Ensuring controlled entry and display of emergency contact persons and contact details.

3.3 Post-Inspection Activity

- Closing meeting to communicate preliminary findings.
- Classification of inspection findings (e.g. critical/ major, intermediate, minor, observations).
- Issuing of an inspection report specifying corrective actions and timelines.
- Follow-up inspections or enforcement actions, where warranted.

4. REQUIREMENTS FOR INSPECTION OF GROUP III HAZARDOUS SUBSTANCES

Inspections of Group III hazardous substances shall assess compliance with the following minimum requirements:

4.1 Licensing and Governance

- Availability of Hazardous Substance Act, 15 of 1973, on site.
- Availability of Regulations R1302 and 1332, on site.
- Valid SAHPRA license covering the specific substances, activities, and locations.
- Appointment and competence of a Radiation Protection Officer or Responsible Person.
- Appointment of Medical Physicist where applicable.
- Availability of a documented radiation protection programme and operating procedures.
- Procedures for incident management and reporting to SAHPRA.

4.2 Radiation Protection and Safety

- Application of the principles of (ALARA), justification, optimisation and dose limitation.

- Adequate control of occupational and public exposure.
- Use of personal dosimetry.
- Workplace monitoring, where applicable.

4.3 Facilities, Storage and Security

- Appropriate shielding.
- Adequate access control, and signage.
- Measures to prevent unauthorised use (in case of mobile units: loss or theft).

4.4 Equipment and Instrumentation

- Availability of suitable radiation monitoring instruments.
- Evidence of calibration.
- Quality Assurance (QA) programme.
- Maintenance and functionality of equipment.

4.5 Records, Incidents, and Reporting

- Records of acquisition, use, transfer, and quality control testing.
- Procedures for incident management and reporting to SAHPRA.
- Occupational dose monitoring records.
- Current proof of professional registration of all practitioners involved.

4.6 Inspection Frequency (Graded Approach – Group III)

Inspection frequency shall be determined based on risk, compliance history, and complexity of the practice, and shall typically be:

- High-risk Group III practices or poor compliance history: at least once every 12 months.
- Moderate-risk Group III practices with satisfactory compliance: at least once every 24 months.
- Low-risk Group III practices with demonstrated sustained compliance: at least once every 36

months.

5. REQUIREMENTS FOR INSPECTION OF GROUP IV HAZARDOUS SUBSTANCES

Inspections of Group IV hazardous substances shall focus on the following minimum requirements:

5.1 Authorization and Governance

- Renewed and current Authority detailing specific substances, activities, locations and covering prescribed conditions.
- Confirmation that the activity falls within the applicable authorization or exemption framework.
- Appointment and competence of a Radiation Protection Officers (RPO and ARPO).
- Appointment of Medical Physicists where required.
- Awareness of Hazardous Substances Act No. 15 of 1973 and Regulations related to Group IV Hazardous Substances, R246 and R247.
- Availability of a documented radiation protection programme and operating procedures.
- Procedures for incident management and reporting to SAHPRA.
- Annually updated practice Internal Rules.
- Evidence of Annual Returns submitted to Radiation Control: radionuclides sub-unit.
- Register or documentation for disposal of radioactive waste.
- Radioactive source register.

5.2 Safe Handling and Use

- Implementation of basic radiation safety practices.
- Adequate supervision and user training appropriate to the practice.
- Implementation of Code of Practice regarding the safe handling and use of Group IV Hazardous Substances.

5.3 Storage and Control

- Appropriate shielding, storage and containment of radioactive substances.
- Measures to prevent loss, damage, or unauthorised access.
- Monthly inventory of all radioactive sources.

5.4 Radiation Awareness and Training

- Evidence of basic radiation safety awareness training for users.
- Availability of written safety instructions.
- Awareness by the RPO and responsible person of regulatory obligations.

5.5 Equipment and Instrumentation

- Availability of suitable radiation monitoring instruments (survey meters, if used on patients, they must be calibrated).
- Personal radiation monitoring dosimeters.
- Radioactive source register (Logbooks).
- In case of medical and research facilities: Hot laboratory availability and functioning (extractor fans, fume hood, laminar flow cabinet, continuous flooring, lead-lined work areas, etc).
- Evidence of Quality Assurance (QA) program, maintenance, and functionality of equipment.

5.6 Records and Incident Management

- Basic inventory and leak tests records.
- Occupational dose monitoring records.
- Proof of professional registration of practitioners
- Reporting and documentation of incidents or abnormal events, where applicable.

5.7 Inspection Frequency (Graded Approach – Group IV)

Inspection frequency shall be determined based on risk, compliance history, and complexity of the practice, and shall typically be:

- Higher-risk Group IV practices or identified compliance concerns: at least once every 12 months.
- Moderate-risk Group IV practices with good compliance history: at least once every 24 - 48 months.
- Very low-risk Group IV practices, inspections conducted on a reactive or sampling basis, or in conjunction with other regulatory activities.

6. REFERENCES

The following related documents are referenced:

- 6.1 Guideline for Code of Practice for Safe Use of Unsealed Radioactive Nuclides, SAHPGL-RDN-RN-06.
- 6.2 INTERNATIONAL ATOMIC ENERGY AGENCY Code of Conduct on the Safety and Security of radioactive sources, 2004
- 6.3 INTERNATIONAL ATOMIC ENERGY AGENCY, Inspection of Radiation Sources and Regulatory Enforcement, IAEA-TECDOC-1526, IAEA, Vienna (2007).
- 6.4 INTERNATIONAL ATOMIC ENERGY AGENCY, Radiation Protection and Safety of Radiation Sources: International Basic Safety Standards, IAEA Safety Standards Series No. GSR Part 3, IAEA, Vienna (2014)
- 6.5 South Africa, 1973. Hazardous Substances Act No. 15 of 1973 in Government Gazette 3834.
- 6.6 South Africa. 1973. Regulations Concerning the Control of Electronic Products (Government Notice R1332 in Government Gazette 3991 of 3 August 1973) issued under the Hazardous Substances Act, 1973 (Act No. 15 of 1973).
- 6.7 South Africa. 1989. Regulation relating to Group III hazardous substances, (Government Notice R 690 in Government Gazette 11823 of 14 April 1989) issued under the Hazardous Substances Act, 1973 (Act No. 15 of 1973).
- 6.8 South Africa. 1991. Hazardous Substances Act 15 of 1973: Regulations: Schedule of listed electronic products (Government Notice R1302 in Government Gazette 13299 of 14 June 1991).
- 6.9 South Africa. 1993. Regulation relating to Group IV Hazardous Substances: Exclusions and Exemptions, (Government Notice R246 in Government Gazette 14596 of 26 February 1993) issued

under the Hazardous Substances Act, 1973 (Act No. 15 of 1973)

- 6.10 South Africa. 1993. Regulations relating to Group IV Hazardous Substances (Government Notice R247 in Government Gazette 14596 of 26 February 1993).

7. VALIDITY

This guideline is valid for a period of 5 years from the effective date of revision. It will be reviewed on this timeframe or as and when required.

APPROVED